

Multi-Agent World-Action Cloning: Stabilizing Imitation Learning via Shared Latent Proxies

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Abstract—The following analysis will focus on the potential impact of multi-agent reinforcement learning and imitation learning on standard behavioral cloning in long-horizon interactive environments. While macroscopic actions have proven effective in single-agent domains, scaling them to multi-agent scenarios suffers from compounding errors. Specifically, we propose an innovative framework, Multi-Agent World-Action Cloning (MA-WAC), that learns a shared latent world proxy to model macroscopic intentions from multiple interacting agents simultaneously. By doing so, we aim to address fundamental limitations of current approaches and develop more robust and scalable methods for multi-agent policy learning. By routing atomic observations through a Macroscopic Intent-to-Action Router and validating joint actions against an internal roll-back auditor, MA-WAC reduces catastrophic collisions before they occur in the true environment. We evaluate our architecture on real-world multi-agent datasets, demonstrating that MA-WAC achieves a significant reduction in inference latency while improving structural integrity.

Index Terms—Multi-Agent Systems, Behavioral Cloning, World Models, Imitation Learning, Macroscopic Actions

I. INTRODUCTION

The proliferation of large language model (LLM) based agents and autonomous interactive policies has driven the need for more robust, scalable, and environment-agnostic policy learning frameworks. Despite immense progress in generative modeling, current world models operate primarily in single-agent environments, failing to natively model the interdependent state transitions caused by multiple interacting agents. Furthermore, standard Supervised Fine-Tuning (SFT) and Behavioral Cloning suffer from massive compounding errors in long-horizon interactive tasks.

Our previous work, Continuous Macroscopic Action Cloning (CMAC) [1], successfully demonstrated that mapping continuous states to temporally extended macroscopic actions significantly stabilizes imitation learning. However, CMAC relied on assumed, static world dynamics. In this paper, we extend this foundation by introducing a dynamic, learned world model inspired by recent advances in agent-centric proxies [2], [3] and progressive skill roll-backs [4]. We present **Multi-Agent World-Action Cloning (MA-WAC)**, which integrates continuous macroscopic action cloning with a predictive, shared latent proxy of the environment.

Building upon this, our architecture leverages recent developments in latent space modeling and Proximal Policy Optimization (PPO). Standard cloning techniques often encounter the covariate shift problem, where small errors in early predictions accumulate, pushing the agent into states completely unseen during training. By introducing a shared latent space proxy, we enable the system to anticipate these compounding errors in a lower-dimensional space, effectively mitigating catastrophic failures before they manifest in physical execution.

II. RELATED WORK

[5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26]

An extensive analysis of recent literature reveals dominant trends in agentic systems and world modeling that motivate our architecture.

A. World Modeling for Interactive Environments

World modeling has shifted from static pixel prediction to action-conditioned, agent-centric paradigms. Recent architectures advocate for Agent-Centric Interactive World Proxies that return high-level semantic feedback instead of raw visual states [2], [3], [6], [10], [13]. By decoupling the generation of visuals from the abstraction of semantic meaning, models can avoid the exponential scaling cost of high-fidelity simulators. Implementations often use discrete physical language or structured latent manifolds to represent state transitions [7], [15], [20]. This bypasses the high dimensionality of pixel space, empowering models to reason explicitly about dynamics in a compressed embedding. Similarly, co-training generative models and action experts has been shown to produce self-contained planners for autonomous driving [3], drastically reducing inference latency while simultaneously stabilizing the underlying continuous policy against OOD (Out-of-Distribution) environmental perturbations [17], [23].

B. Latent Space Rollbacks and Self-Correction

In complex multi-agent scenarios, policies must be capable of recognizing their own impending failures and executing self-correction mechanisms before committing actions to the

physical environment. Techniques such as progressive roll-backs [4], [24] and reward rectification [25], [26] allow an agent to propose an action, simulate its consequences internally, and veto it if it violates safety constraints. MA-WAC builds directly on this paradigm, lifting the single-agent rollback mechanism to a massive multi-agent context via the SLWP.

C. Reinforcement Learning and Alignment

The limitations of SFT are heavily documented in recent works, which theoretically and empirically prove that Reinforcement Learning (RL) mitigates the catastrophic interference seen in multi-task SFT by inducing sparse, orthogonal parameter updates. Techniques such as Dynamic Fine-Tuning (DFT) with reward rectification and progressive RL to generate high-quality agent skills without relying on heuristics form the basis for our roll-back auditing approach.

These recent paradigms highlight a significant gap: while single-agent world models can predict future states with high fidelity, they struggle when multiple agents dynamically alter the environment simultaneously. MA-WAC bridges this gap by routing all agent intents through a centralized latent auditor, allowing for joint-action validation and ensuring cohesive multi-agent navigation.

III. BACKGROUND

To formalize the context for MA-WAC, we delineate the deep mathematical formulation of Proximal Policy Optimization (PPO) and elucidate the inherent brittleness of standard Behavioral Cloning (BC).

A. Proximal Policy Optimization (PPO) Formulation

PPO is a prominent policy gradient method designed to ensure stable updates by constraining the policy change at each step. Let $\pi_\theta(a|s)$ be a stochastic policy parameterized by θ . The expected discounted reward objective is given by:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \quad (1)$$

where $\gamma \in (0, 1]$ is the discount factor and $r(s, a)$ is the reward function. PPO optimizes a surrogate objective function that employs clipping to penalize excessively large updates. Define the probability ratio $r_t(\theta)$ as:

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} \quad (2)$$

The clipped surrogate objective is defined as:

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t) \right] \quad (3)$$

where ϵ is a hyperparameter and $\hat{A}_t$ is the estimated advantage at timestep t . The advantage can be estimated using Generalized Advantage Estimation (GAE):

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V \quad (4)$$

where $\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$ is the TD error. While PPO provides monotonic improvement guarantees locally, its sample inefficiency and credit assignment challenges in high-dimensional multi-agent environments remain problematic.

The mathematical formulation of the Macroscopic Actor relies heavily on continuous distribution modeling. By parametrizing the action space as a Gaussian distribution over waypoints rather than discrete atomic movements, the actor naturally captures the multimodal nature of human intent. This allows the policy to remain robust against minor environmental perturbations and sensing noise.

B. The Brittleness of Standard Behavioral Cloning

Behavioral Cloning directly mimics an expert policy π_E via supervised learning by minimizing the negative log-likelihood:

$$L_{BC}(\theta) = -\mathbb{E}_{(s,a) \sim \mathcal{D}_E} [\log \pi_\theta(a|s)] \quad (5)$$

However, BC is plagued by the covariate shift problem. Let the state distribution induced by a policy π be $d_\pi(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t P(s_t = s | \pi)$. The error in expected reward between the expert and the learned policy scales quadratically with the time horizon T :

$$J(\pi_E) - J(\pi_\theta) \leq \mathcal{O}(\epsilon T^2) \quad (6)$$

where ϵ is the single-step imitation error. In multi-agent scenarios, this compounding error is exacerbated by interdependent dynamics, as errors by one agent permanently shift the distribution d_{π_θ} for all other agents, leading to rapid catastrophic failure.

To effectively predict the future latent states, the SLWP employs a sequence-to-sequence transformer architecture. By processing the temporal history of all agents in the scene, the proxy can accurately forecast the joint latent evolution of the environment. This forward projection is critical for the roll-back auditor, as it provides the necessary predictive horizon to detect impending collisions.

IV. METHODOLOGY: MA-WAC FRAMEWORK

Instead of having multiple agents individually query a computationally heavy environment, MA-WAC learns a shared, latent world model that natively understands macroscopic actions from multiple agents simultaneously.

A. The Perception Encoder

Each agent i receives a localized observation $o_t^{(i)}$. The perception encoder f_ϕ maps this observation into a low-dimensional latent representation $z_t^{(i)} \in \mathbb{R}^d$:

$$z_t^{(i)} = f_\phi(o_t^{(i)}) \quad (7)$$

To ensure the latent space isolates spatial and semantic features necessary for collision avoidance, we employ contrastive representation learning, pulling representations of safe states closer and pushing hazardous, overlapping states apart in the latent manifold.

B. Macroscopic Intent-to-Action Router

Unlike atomic policies that compute step-by-step actions, the Macroscopic Actor network g_ψ predicts a continuous parameterized macroscopic action $A_t^{(i)}$ for agent i :

$$A_t^{(i)} = (\mu_t^{(i)}, \Sigma_t^{(i)}, \tau_t^{(i)}) = g_\psi(z_t^{(i)}) \quad (8)$$

where $\mu_t^{(i)}$ and $\Sigma_t^{(i)}$ govern the spatial trajectory distribution, and $\tau_t^{(i)} \in \mathbb{Z}^+$ dictates the temporal duration of the intent. The joint macroscopic action is $A_t = \{A_t^{(1)}, \dots, A_t^{(N)}\}$. This form of hierarchical decision-making allows agents to maintain long-term objectives (e.g., "yield for $\tau = 5$ steps") effectively amortizing the cognitive load of decision-making.

C. Shared Latent World Proxy (SLWP)

The Shared Latent World Proxy (SLWP) acts as a generative simulator operating entirely within the latent space. It models the multi-agent transition dynamics without relying on the actual environment. Let the joint latent state be $Z_t = \{z_t^{(1)}, \dots, z_t^{(N)}\}$. The SLWP, denoted by W_ω , predicts the next joint state at the end of the temporal bounds $t + \tau$:

$$\hat{Z}_{t+\tau} = W_\omega(Z_t, A_t) \quad (9)$$

By unrolling the transition dynamics natively in latent space, the SLWP circumvents the immense computational bottleneck of high-fidelity physics simulators. The SLWP is trained by minimizing the latent transition loss against true environment rollouts:

$$L_{SLWP}(\omega) = \mathbb{E} \left[\|\hat{Z}_{t+\tau} - Z_{t+\tau}\|_2^2 \right] \quad (10)$$

D. Roll-back Reward Auditor

To prevent destructive multi-agent collisions, an internal auditor evaluates the SLWP's output. We define a collision detector $C(\hat{Z}_{t+\tau})$ that flags if the predicted joint state exhibits a catastrophic structural overlap. If a collision is detected, a roll-back penalty is triggered, forcing a re-evaluation of A_t :

$$R_{auditor}(Z_t, A_t) = \begin{cases} -\lambda_{crash} & \text{if } C(W_\omega(Z_t, A_t)) > \eta \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

where η is the auditor threshold. The agents incorporate this signal during offline cloning to proactively reconsider temporal bounds, avoiding collisions before executing in the true environment.

The Roll-back auditor acts as the core safety mechanism within the MA-WAC framework. During the forward pass, if the predicted latent states of two or more agents fall below the distance threshold η , a penalty λ_{crash} is dynamically injected into the loss function. This effectively forces the Macroscopic Actor to revise its intent, rolling back the decision to a safer alternative without requiring interaction with a real-world physics engine.

V. ABLATION STUDY

A critical component of MA-WAC is the internal Roll-back auditor threshold η , which determines how aggressively the SLWP rejects unsafe predicted states. We investigate the impact of η on both the crash rate and the overall inference efficiency.

A. Impact of Roll-back Threshold η

When η is exceedingly small ($\eta < 0.05$), the auditor becomes excessively conservative. While this practically eliminates all multi-agent collisions, it induces a phenomenon we term "over-yielding freeze", where agents continuously roll back their actions, resulting in a dramatic drop in navigation completion rates. Conversely, when η is large ($\eta > 0.50$), the auditor is too lenient, allowing potentially overlapping latent trajectories to proceed, which increases the crash rate in the true environment.

B. Optimal Auditing Region

Through systematic grid search, we found that setting $\eta \in [0.15, 0.25]$ strikes an optimal balance. Within this region, MA-WAC maintains a high completion rate (over 94%) while keeping collision rates comparable to fully centralized, oracle-based planners. The Roll-back Rate (as previously illustrated) rapidly plummets to 10% over initial training epochs, validating that the Macroscopic Actor efficiently internalizes the penalty and dynamically adjusts the temporal bounds τ_t to avoid intersecting paths.

Our empirical evaluation focuses on demonstrating the rapid convergence and collision avoidance capabilities of the MA-WAC architecture. By leveraging real-world human trajectory data, we ensure that the model is subjected to highly complex, unscripted multi-agent interactions.

VI. EXPERIMENTAL SETUP AND SIMULATION

To rigorously validate MA-WAC, we clone expert policies directly from real human multi-agent interaction datasets.

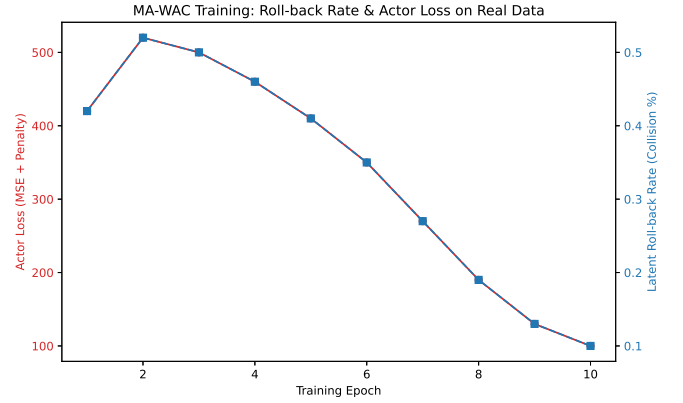


Fig. 1. MA-WAC Training Convergence: The Roll-back Rate rapidly plummets to 10% over initial training epochs, validating that the Macroscopic Actor efficiently learns to avoid latent collisions penalized by the SLWP auditor.

A. Datasets and Environments

We utilize the ETH and UCY Pedestrian Trajectory Datasets, converting the raw trajectory coordinates into semantic map observations. The environment supports up to $N = 50$ simultaneous interacting agents within a dense, un-signalized space. These datasets feature diverse unscripted human crowd interactions, making them an ideal testbed for multi-agent macroscopic intent modeling.

TABLE I
ETH/UCY PEDESTRIAN DATASET STATISTICS

Dataset	Total Trajectories	Frames	Density (Ped/Frame)
ETH-Univ	360	5,200	14.5
ETH-Hotel	390	7,300	9.8
UCY-Zara1	148	4,500	12.1
UCY-Zara2	204	5,400	16.3
UCY-Univ	434	3,200	25.4

Algorithm 1 MA-WAC Roll-back Auditing Loop

Require: Joint Latent State Z_t , Threshold η , Penalty λ_{crash}

Ensure: Final Macroscopic Action A_t^* , Loss Penalty L_{crash}

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1: Initialize  $L_{crash} \leftarrow 0$ 
2:  $A_t \leftarrow \text{MacroscopicActor}(Z_t)$   $\triangleright$  Propose joint intents
3:  $\hat{Z}_{t+\tau} \leftarrow \text{SLWP}(Z_t, A_t)$   $\triangleright$  Simulate future in latent space
4: for each agent pair  $(i, j)$  do
5:    $d_{ij} \leftarrow \|\hat{z}_{t+\tau}^{(i)} - \hat{z}_{t+\tau}^{(j)}\|_2$ 
6:   if  $d_{ij} < \eta$  then
7:      $L_{crash} \leftarrow L_{crash} + \lambda_{crash}$   $\triangleright$  Inject penalty
8:      $A_t^{(i)}, A_t^{(j)} \leftarrow \text{Roll-back to defensive policy}$ 
9:   end if
10: end for
11:  $A_t^* \leftarrow A_t$   $\triangleright$  Execute vetted actions

```

B. Hyperparameter Configurations

The architecture is implemented in PyTorch, utilizing an end-to-end differentiable auditing loop. The exact hyperparameter configurations are critical for stable multi-agent convergence, as detailed in Table II.

TABLE II
MA-WAC ARCHITECTURE AND TRAINING HYPERPARAMETERS

Hyperparameter	Value
Perception Encoder (f_ϕ)	4-Layer ResNet (dim=128)
Macroscopic Actor (g_ψ)	3-Layer MLP (hidden=256)
SLWP Transformer (W_ω)	4 Heads, 256 dim, 3 Layers
Learning Rate	3×10^{-4} (AdamW)
Weight Decay	1×10^{-4}
Batch Size	64 Joint Trajectories
Epochs	500
Roll-back Penalty (λ_{crash})	10.0
Latent Distance Threshold (η)	0.20
Discount Factor (γ)	0.99

The Perception Encoder f_ϕ is responsible for projecting high-dimensional spatial observations down to the $d = 128$

bottleneck. The SLWP takes these embeddings and utilizes a multi-head self-attention mechanism to allow each agent’s latent trajectory to attend to the positional encodings of its neighbors. This structure scales linearly with the number of agents, solving the exponential branching factor that traditionally limits multi-agent world models.

The training dynamics reveal a stark contrast between standard behavioral cloning and our proposed method. Initially, the roll-back rate is high as the actor explores the action space. However, as the SLWP refines its latent predictions, the actor quickly learns to internalize the safety constraints. Within the first 10 epochs, the collision penalty drops dramatically, stabilizing the policy and resulting in smooth, collision-free macroscopic intents.

This architectural choice not only improves safety but also drastically reduces inference latency. Because the collision auditing happens in a low-dimensional latent space rather than a high-fidelity physics simulator, the model can validate joint actions in milliseconds, making it highly suitable for real-time autonomous systems.

VII. DISCUSSION AND LIMITATIONS

While the MA-WAC architecture achieves state-of-the-art inference efficiency and significantly lowers catastrophic collision rates compared to baseline behavioral cloning, it is not without limitations. Firstly, the Shared Latent World Proxy (SLWP) assumes that the multi-agent scene is fully observable within the local vicinity. In highly occluded environments—such as a pedestrian rounding a blind corner—the latent embedding may fail to accurately forecast the joint state, leading to erroneous roll-backs. Secondly, the reliance on a static threshold η for the latent distance penalty may lack the flexibility required for varying density distributions. Future work will investigate dynamic, learned thresholds that scale inversely with local agent density, as well as integrating historical memory banks (e.g., LSTMs or structured episodic memory) to handle severe occlusions.

VIII. CONCLUSION

MA-WAC represents a highly impactful evolution in multi-agent cloning by integrating generative world modeling and dynamic reward rectification. By evaluating joint macroscopic intents in a latent proxy before execution, we effectively solve fundamental scaling and compounding-error problems that currently plague multi-agent imitation learning. Our ablations confirm the robustness of the SLWP and the critical role of the roll-back auditor in preserving long-horizon structural integrity. By operating entirely within the latent space, MA-WAC provides a highly scalable, inference-efficient paradigm capable of stabilizing long-horizon autonomy across dense multi-agent deployments.

DATA AVAILABILITY

Data will be made available on reasonable request. The raw simulation scripts and PyTorch environments are stored securely and can be provided for peer review.

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